RETAIL SALES & CUSTOMER BEHAVIOUR ANALYTICS

Dataset Documentation & Data Dictionary

Project Overview

This fictional multi-city retail dataset is intended for a practical Data Science project. Students will explore, clean, filter, aggregate, join, and visualise the supplied data using Pandas, Matplotlib, and Seaborn.

Dataset Files

CSV file	Rows	Columns	Purpose
customers.csv	81	9	Customer profile and registration information.
orders.csv	210	7	Order-level information, including date, channel, status, shipping, and payment method.
order_items.csv	525	7	Line-item details. An order may contain multiple product rows.
products.csv	36	7	Product catalogue with category, brand, selling price, cost, and availability status.
payments.csv	167	6	Payment records linked to orders. Payment rows may be missing for some orders.
returns.csv	19	5	Product-return records for orders marked as returned.

Table Relationships

• customers.customer_id → orders.customer_id (one customer can place many orders).

• orders.order_id → order_items.order_id (one order can contain many line items).

• products.product_id → order_items.product_id (one product can appear in many line items).

• orders.order_id → payments.order_id (an order may have a related payment record).

• orders.order_id → returns.order_id (an order may have a return record).

Table-by-Table Data Dictionary

1. CUSTOMERS

Customer profile and registration information. **Rows:** 81. **Columns:** 9.

Column	Description
customer_id	Unique identifier for each customer; primary key.
customer_name	Customer's display name.
gender	Customer gender category.
age	Customer age in years; contains intentionally missing values.
city	Customer's city; contains intentionally missing values.
state	State associated with the customer.
customer_segment	Customer segment: Consumer, Corporate, or Small Business.
signup_date	Date on which the customer registered.
preferred_contact	Preferred communication channel: Email, SMS, or WhatsApp; may be missing.

2. ORDERS

Order-level information, including date, channel, status, shipping, and payment method. **Rows:** 210. **Columns:** 7.

Column	Description
order_id	Unique identifier for each order; primary key.
customer_id	Customer who placed the order; foreign key to customers.customer_id.
order_date	Date on which the order was placed.
sales_channel	Channel used: Website, Mobile App, or Store.
order_status	Order state: Delivered, Shipped, Cancelled, or Returned.
shipping_type	Shipping option: Standard, Express, or Same Day; may be missing.
payment_method	Method selected for payment: COD, Card, UPI, or Net Banking.

3. ORDER_ITEMS

Line-item details. An order may contain multiple product rows. **Rows:** 525. **Columns:** 7.

Column	Description
item_id	Unique identifier for each order-item row; primary key.
order_id	Order containing the item; foreign key to orders.order_id.
product_id	Product purchased; foreign key to products.product_id (one intentional unmatched key is included).
quantity	Number of units in the line item.
unit_price	Selling price per unit recorded for the line item.
discount_pct	Discount rate expressed as a decimal (for example, 0.10 means 10%); may be missing.
line_total	Final line amount after the recorded discount.

4. PRODUCTS

Product catalogue with category, brand, selling price, cost, and availability status. **Rows:** 36. **Columns:** 7.

Column	Description
product_id	Unique identifier for each product; primary key.
product_name	Product name.
category	Product category, such as Electronics, Furniture, Clothing, Beauty, Sports, or Books.
unit_price	Catalogue selling price.
unit_cost	Estimated product cost.
brand	Brand label.
product_status	Product availability status: Active or Discontinued.

5. PAYMENTS

Payment records linked to orders. Payment rows may be missing for some orders. **Rows:** 167. **Columns:** 6.

Column	Description
payment_id	Unique identifier for each payment record; primary key.
order_id	Order associated with the payment; foreign key to orders.order_id (one intentional unmatched key is included).
amount	Payment amount recorded.
payment_status	Payment outcome: Success, Failed, or Pending; may be missing.
payment_mode	Payment mode recorded for the payment.
payment_date	Date on which the payment was recorded; may be missing.

6. RETURNS

Product-return records for orders marked as returned. **Rows:** 19. **Columns:** 5.

Column	Description
return_id	Unique identifier for each return; primary key.
order_id	Order associated with the return; foreign key to orders.order_id.
return_reason	Reason for return, such as Damaged, Wrong Item, Size Issue, Late Delivery, or Changed Mind; may be missing.
days_to_return	Number of days between delivery/purchase and return as represented in this practice dataset.
refund_status	Refund workflow status: Approved, Rejected, or Under Review.

Data Quality Notes

• Missing values are intentionally included in selected columns.

• The customers table contains one duplicated customer record.

• The order_items table contains an intentionally unmatched product_id.

• The payments table contains an intentionally unmatched order_id.

• Preserve raw DataFrames and create separate cleaned copies.

• Convert date columns to datetime before date-based analysis.

• State how cancelled and returned orders are treated in revenue calculations.

Suggested CSV Loading Code

```
import pandas as pd
customers = pd.read_csv('customers.csv')
orders = pd.read_csv('orders.csv')
order_items = pd.read_csv('order_items.csv')
products = pd.read_csv('products.csv')
payments = pd.read_csv('payments.csv')
returns = pd.read_csv('returns.csv')
```